សាកលវិទ្យាល័យភ្នំពេញអន្តរជាតិ

PHNOM PENH INTERNATIONAL UNIVERSITY

Machine Learning

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FINAL PROJECT

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SMART VEHICLE ACCESS · COMPUTER VISION

License Plate Detection Using Machine Learning

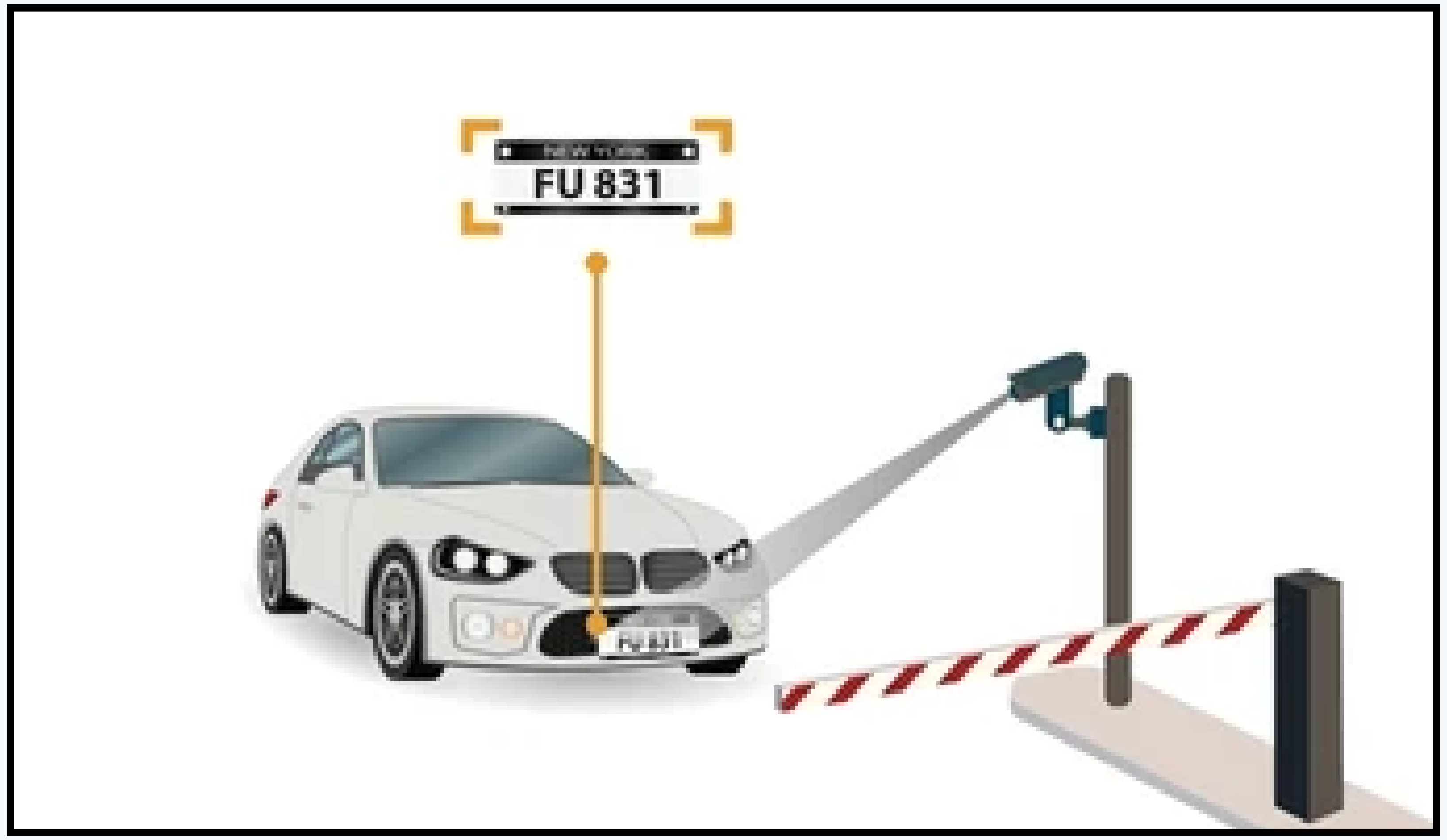
Smart License Plate Recognition & Auto Gate System

Table Content

What Is License Plate Detection?

A computer-vision technique

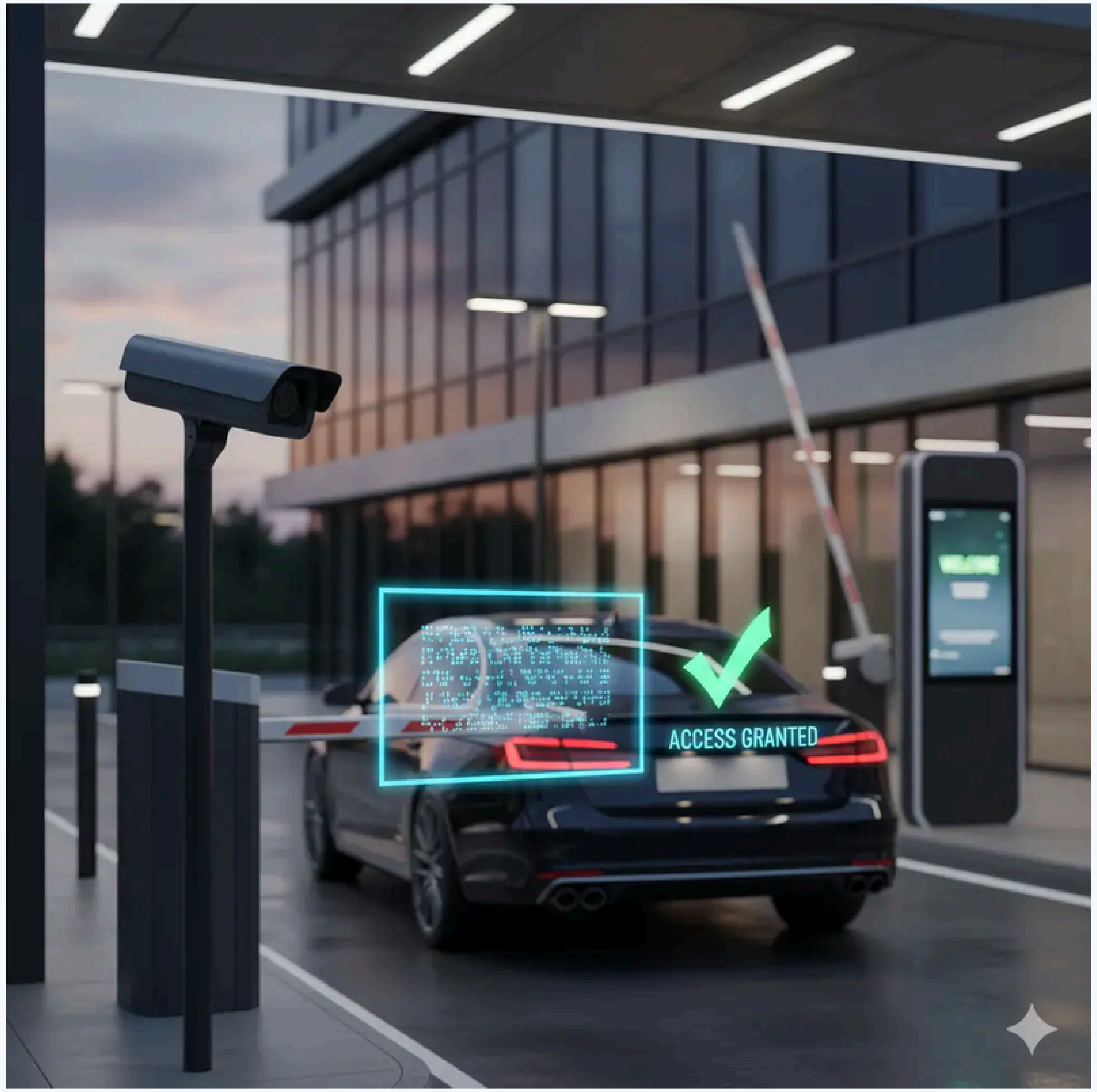
License plate detection automatically locates a vehicle's license plate in an image or video the first step toward reading and verifying it.



What Is License Plate Detection?

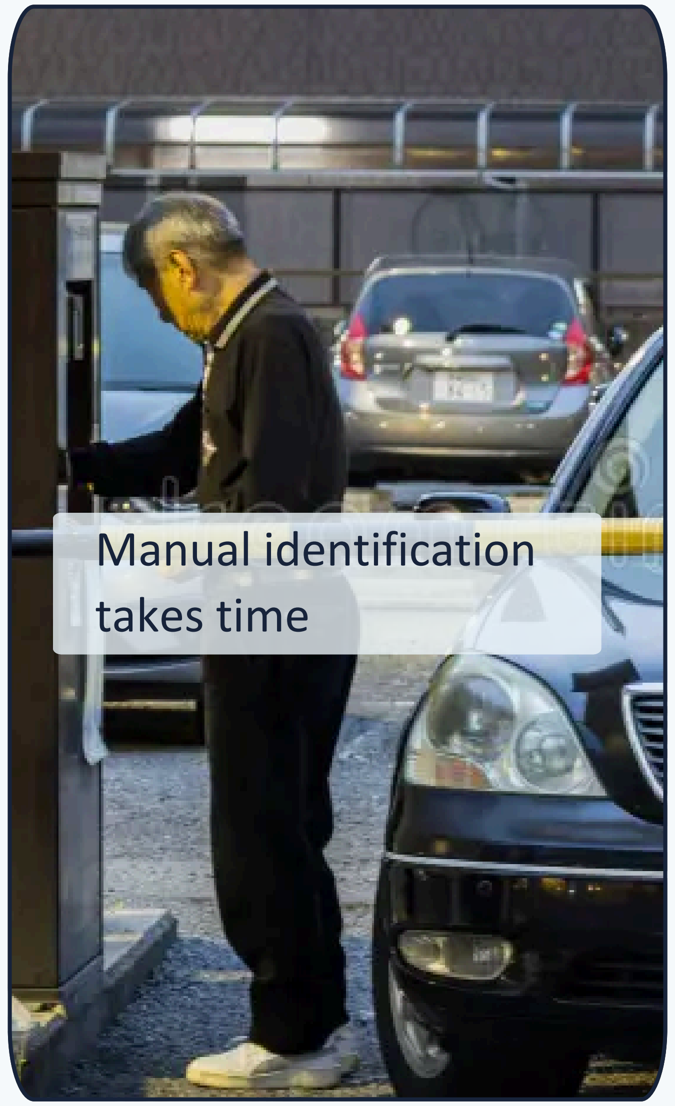
This Project

The system detects a license plate using YOLOv8, reads the characters using OCR, and checks the plate against an authorized database.

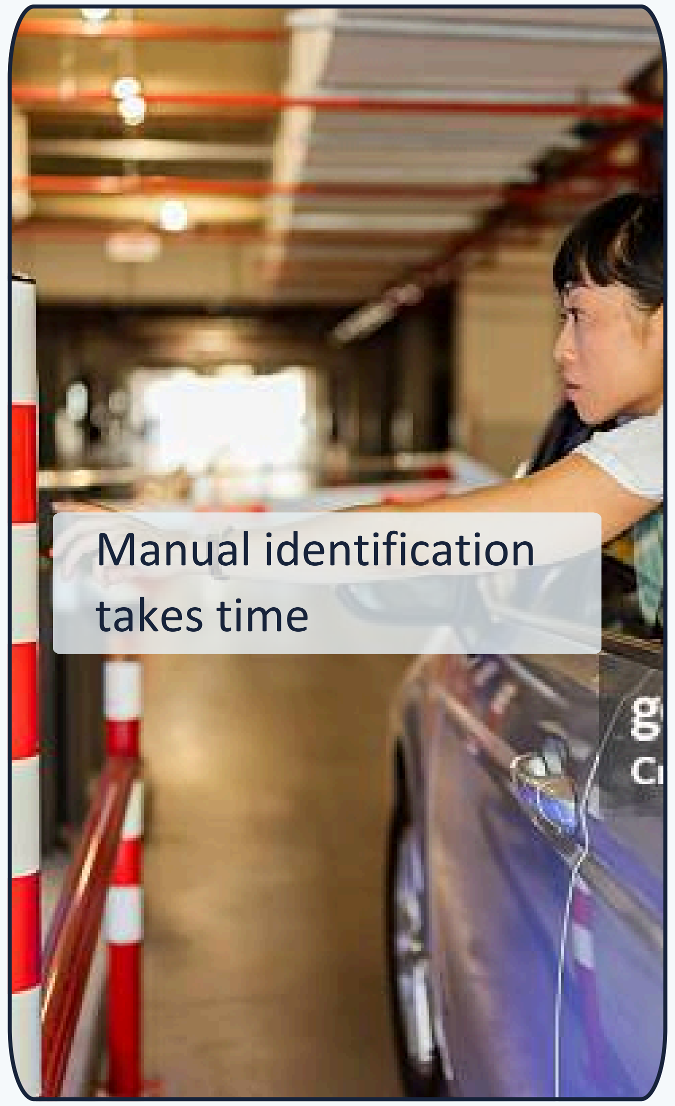


Manual Access Control Doesn't Scale

Traditional vehicle access systems require people to manually check license plates — slow, inconsistent, and error-prone.


Manual identification takes time

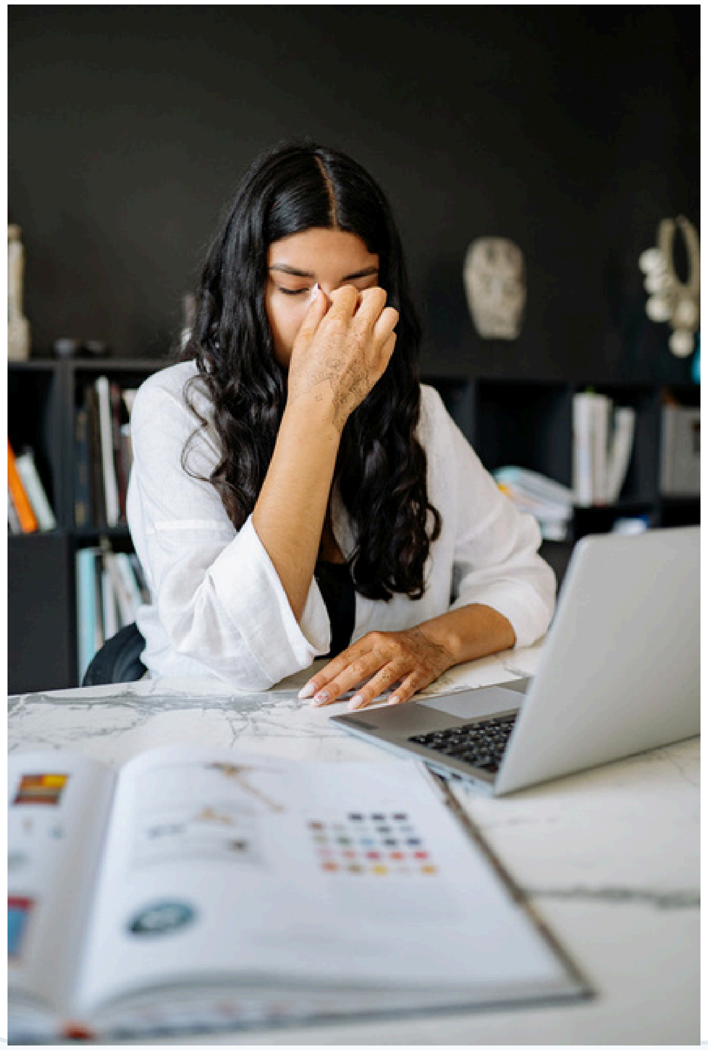
Human operators can make mistakes


Manual identification takes time


Access records are hard to maintain

An automatic system that can Detect → Recognize → Verify → Control access.

MANUAL PROCESS



Vehicle arrives → Human reads plate → Check list → Open / Deny

An automatic system that can Detect → Recognize → Verify → Control access.

MANUAL PROCESS

Vehicle → AI Detection → OCR → Database

Five Goals, One Pipeline

01 Detect

Automatically detect license plates from vehicle images.

02 Recognize

Use OCR to convert the plate image into text.

03 Verify

Compare the recognized plate number with the authorized database.

04 Control

Automatically open the gate for authorized vehicles.

05 Record

Store the access event in the database.

Machine Learning Background

Machine Learning (Supervised Learning)

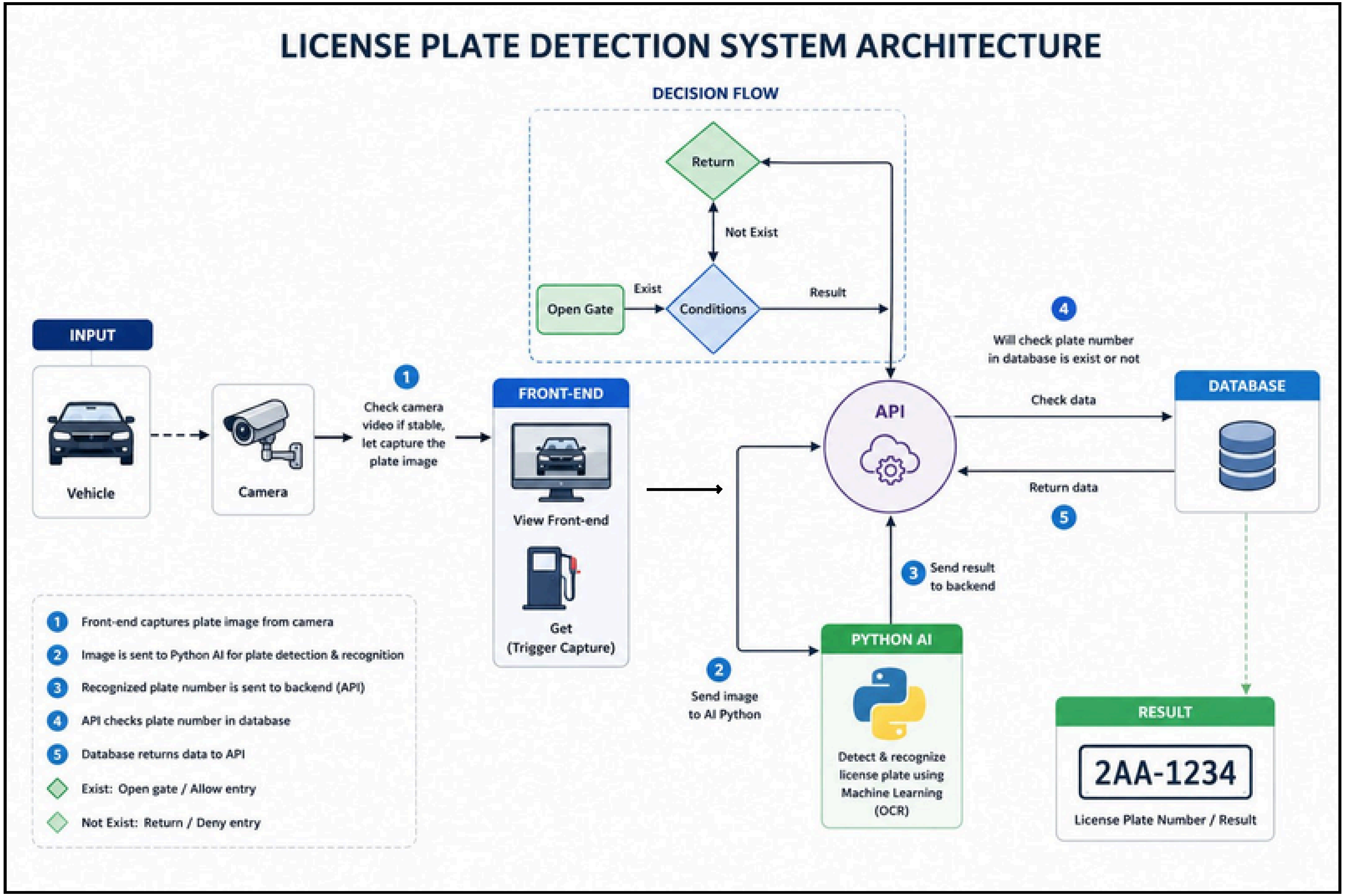
Allows computers to learn patterns from data and make predictions without manually programming every rule.

Object Detection -YOLOv8

YOLO identifies where the license plate is: the bounding-box coordinates plus a detection confidence score.

OCR

Optical Character Recognition converts an image containing characters into text. This project uses EasyOCR.


LICENSE PLATE DETECTION SYSTEM ARCHITECTURE
DECISION FLOW
Return
Not Exist
Exist
Open Gate
Conditions
Result
INPUT
Vehicle
Camera
1
Check camera video if stable, let capture the plate image
FRONT-END
View Front-end
Get (Trigger Capture)
API
4
Will check plate number in database is exist or not
Check data
Return data
5
DATABASE
3
Send result to backend
2
Send image to AI Python
PYTHON AI
Detect & recognize license plate using Machine Learning (OCR)
RESULT
2AA-1234
License Plate Number / Result
1 Front-end captures plate image from camera
2 Image is sent to Python AI for plate detection & recognition
3 Recognized plate number is sent to backend (API)
4 API checks plate number in database
5 Database returns data to API
Exist: Open gate / Allow entry
Not Exist: Return / Deny entry

Teaching the Model to See Plates

TRAINING — 50 EPOCHS

The training curves show consistent, healthy convergence:

Box loss	Decreased steadily
Classification loss	Decreased steadily
DFL loss	Decreased steadily
Validation losses	Generally decreased
Precision & recall	Improved over training

Teaching the Model to See Plates

4,563
Image

Roboflow
Website

50
Epoch

4
Train

Detection Engine: YOLOv8 (Ultralytics)

WHY YOLO?

- Fast detection
- Good for real-time applications
- Supports custom datasets
- Produces bounding boxes
- Easy Python integration

MODEL STORAGE PATH

```
models/
└─ plate_detection/
   └─ runs/
      └─ detect/
         └─ train/
            └─ weights/
               └─ best.pt
```

IN YOUR SYSTEM

```
model = YOLO("best.pt")
results = model(image, conf=0.5)
```

Returns: Bounding Box + Confidence

Model Performance

Precision ≈ 91%

When the model detects a plate, it is usually correct.

Recall ≈ 81%

The model successfully finds most plates in the validation data.

mAP@50 ≈ 85%

Strong performance under the IoU 0.50 evaluation threshold.

mAP@50–95 ≈ 65%

Detection is harder when stricter bounding-box accuracy is required.

Real Dashboard Result

TEST RESULT

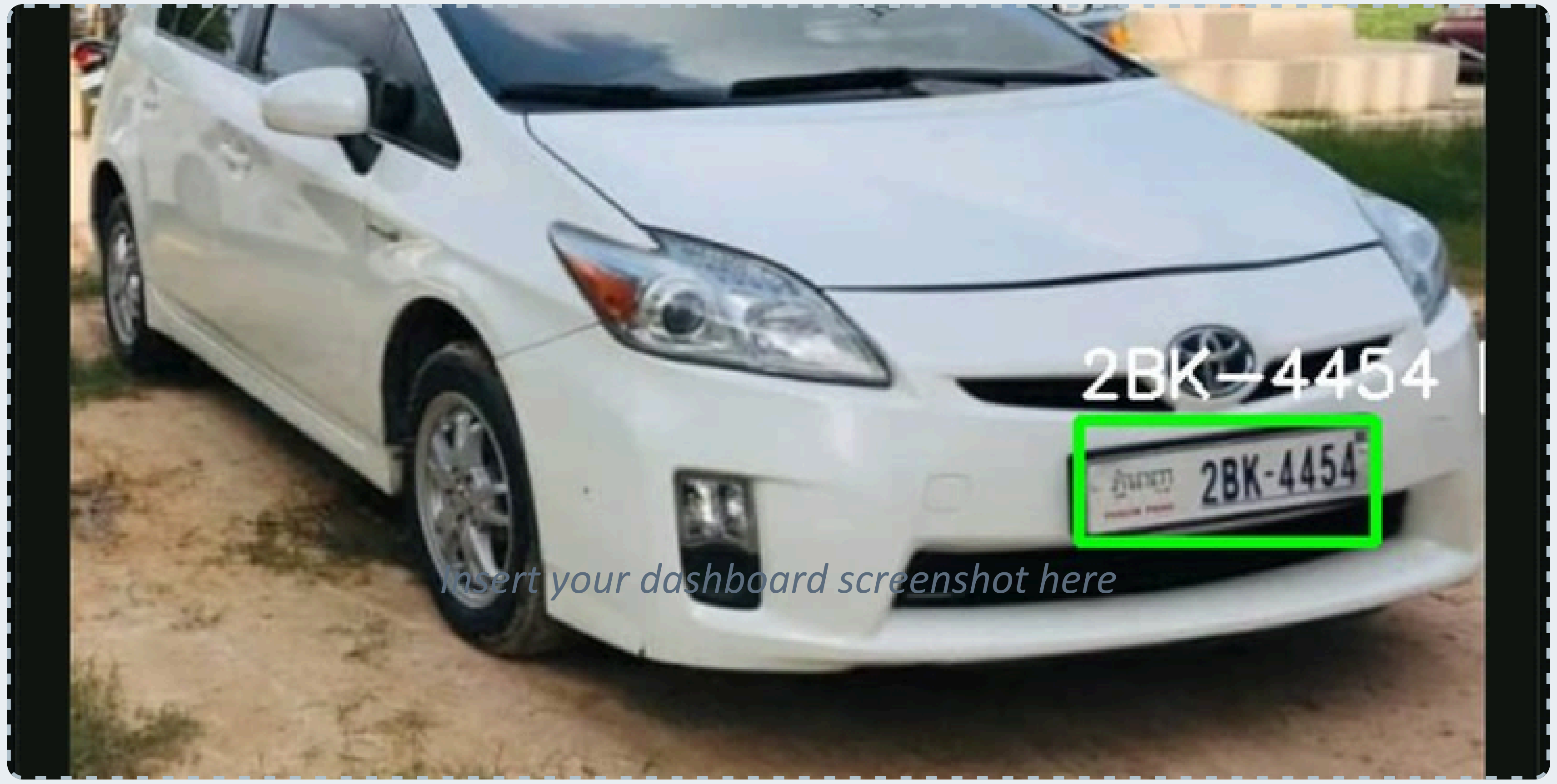
Vehicle Image (Input)

YOLO — License plate detected

OCR — 2BK-4454

Database — Authorized

✅ GATE OPENED



Why this matters

Plate 2BK-4454 was recognized, authorized, and the gate opened — proving the project is not just a trained model, but a working integrated application.

What We Built

A Smart License Plate Recognition system that combines:

YOLO + OCR + Python + Flask API + Database + Gate Controller

Main achievement

Detect → Read → Verify → Open / Deny → Log

85%
MAP@50

91%
PRECISION

81%
RECALL

This project demonstrates how Machine Learning and OCR can be integrated into a practical, automated vehicle access-control system.

Thank You

Questions?

Smart License Plate Recognition & Auto Gate System